

Project #10

See course website for due date.

You are encouraged to comment directly on this document.

Overall Project Goal:

The plan is to create a car that can follow a black electrical tape line after navigating via a Wi-Fi connection. The creation of the car is broken into ten projects.

The first project is building the battery power system. Verification will be performed using Either Analog Discovery or other Lab test equipment.

The second project is the installation of the LCD onto the Control Board and the material to attach it to the FRAM board. Demo your working LCD with the Project 2 code.

The third project is the creation of the car with the Forward only part of the H-bridge control for the wheels. Successful verification will be movement of the car and appropriate car labeling.

The fourth project is the controlled movement of the car with the forward only part of the H-bridge control for the wheels. Successful verification will be of the car moving in predetermined shapes within a 36" x 36" square area.

The fifth project is adding full control of the H-Bridge allowing the vehicle to travel forward and reverse. Verification will be with timed travel forward followed by timed travel in reverse; ending with spinning motion.

The sixth project is the addition of the emitter / detector circuit that will detect the black line, and the code to resolve the Thumbwheel to a digital value to verify ADC configuration. Verification will be manual display of ADC values as your vehicle transitions from white to black and black to white, independent tracking of the Thumbwheel. Your vehicle must travel from inside the 36-inch Diameter circle and stop when it detects the circle. After circle detection, your car must turn approximately 90 degrees and be detecting the black line.

The seventh project will be adding traveling around the circle two or more times and an exit into the middle of the circle after the second time around.

The eight project is serial communication. There will be homework assignments that will assist in building software routines that can be used in the project.

The ninth project is to communicate with a Wi-Fi device.

The tenth project is composed of two distinct portions. First, you must navigate your car around a course in the atrium of EBII remotely using a Wi-Fi connection. Then, you will give your car a single command that will tell it to go find the black line. It will then autonomously find the line

and follow it until directed otherwise. Bonus points will be awarded to those demoing to full completion on the first attempt.

The Wi-Fi course consists of driving around to eight numbered pads that are placed around the EBII atrium. At each pad, you must drive onto the pad and stop. Then you must send a command to display “Arrived 0X” on the top line of your display, where X is the number of the pad you are on. There will be a TA or designated assistant at each of the numbered pads. Once you have arrived on the pad and your display is showing “Arrived 0X”, you must verbally confirm with the assistant that they successfully entered you into their grade logbook for completion of that pad.

The eight pad is different from the rest. This pad signifies the end of the IOT course and the beginning of the autonomous line finding and following portion of the course. There are special rules about how you must control your car around this pad, so be sure to pay special attention to those.

Once you are within two feet of pad 8, a 45-second timer is started. Within those 45 seconds, you must approach pad 8 and touch at least two wheels onto it. Once two wheels have touched the pad, you can only send one command to make your car autonomously find and follow the line. (It is permissible to send a single command prior to arriving on pad 8 that drives completely over pad 8, and *then* issue your final command, as that still counts as sending only one command after two wheels have touched the pad). This command must be issued within the 45 second window.

The Caster arrangement counts as a wheel. If you touch with a wheel and a “Caster”, you have satisfied the “two wheels have touched the pad” requirement.

After that, your car must autonomously navigate from the eight pad to the board with the black line and circle on it. It must detect the line, turn onto the line, and follow it down to the circle and around the circle twice. During this time, you may not send any commands. Only after your car has traveled around the circle at least twice and you have received confirmation from the TA, you may issue *one* final command. This command will cause the vehicle to stop, turn facing away from the circle, drive at least two feet away from the circle, and stop. You must display a message indicating successful completion.

Project 10 Directions:

In this project you will write software and use your previously assembled hardware to make your car navigate a course using the IOT Module and then transition to intercepting and following a black line and circle. The work on the project can be a collective effort between team members, but you may not share code.

The goal of Project 10 is to put all the individual pieces of code you have made over the semester together. This can cause issues such as code limits, insufficient power supply issues, etc., so be sure to allocate sufficient time to deal with these roadblocks.

The IOT portion of the course will require you to navigate around some obstacles of furniture. None of the numbered square pads are affixed to the floor, so they may move from their initial locations due to student vehicle operation. Re-placement will not happen unless the number is

moved more than a few feet away from its original location, and only TA's or the Instructor can reposition the pads.

To start the IOT course, you will wait in line near pad 0 and sign in on the attendance sheet. Then, when it is your turn you must first show that your car is off before turning it on in front of the Instruction/Attendant. It must automatically connect to Wi-Fi by itself (i.e. without a serial connection to Termite), and it must display the IP address and SSID to demonstrate successful connection to the Wi-Fi.

At this point, you may calibrate your car for the black and white values in the current ambient lighting. There will be an extra board near the start of the course where you can complete this calibration, and once the calibration is finished your display will say "Waiting for input".

Then, you will put your car down on or near pad 0 (this is the only pad where it is permissible to not be exactly *on* the pad) and start the course, navigating to each numbered pad in sequence. As soon as the first command is received, your car must start a second's counter that displays the current time the course has taken, in seconds.

Successful arrival at a pad is defined as two wheels touching the pad, with the TA verifying successful completion of each pad. If you have any issues during the IOT course before reaching pad 8, or you must touch your car for any reason, you must restart the entire course at pad 0.

While traversing the course, you must stay clear of the main course area and watch / control from the sidelines of the course. This is so no cars are stepped on by you or passersby. If you must pass through the course for any reason (e.g. to get to and from the lab for testing, or to have a better vantage point from which to operate your car), Alert one of the TAs/Attendants before walking slowly and carefully across the course while avoiding all cars by a wide margin. It is preferred that you go outside and around to the rear of EBII rather than going through the Atrium.

The course will have several cars operating at the same time. Accidents and other obstructions are very likely, so make sure your car and your controlling device have commands to stop your car in an emergency. This will help reduce the risk to your car or other's.

Ordinarily, there is a queue of cars right before pad 7, as the remaining portion of the course (from pad 7 to pad 8 and on to the black line) can only be attempted by one student at a time. You will wait in line before or on pad 7, and then once the TA has confirmed they are ready for you, you will travel to pad 8.

Once two wheels touch pad 8, that counts as successful completion of the IOT course, so if there is an error after that point you do not have to restart from the very beginning of the IOT course, you just must go back to the queue before pad 7 and start from there.

Priority in the queue before pad 7 is given to students who are completing the IOT course (AKA, those who have not attempted the black line portion yet). If you are attempting the black line portion for any attempt after your first, you must yield to students finishing up the IOT portion of the course coming from pad 6.

Pad 7 will be placed at least six feet from pad 8. All of the pads are one foot square. There will be two white panels with identical black line shapes on them: the first 2 feet to the right of pad 8 and the other two feet to the left. Please see Figure 1.1 for reference of what this orientation looks like.

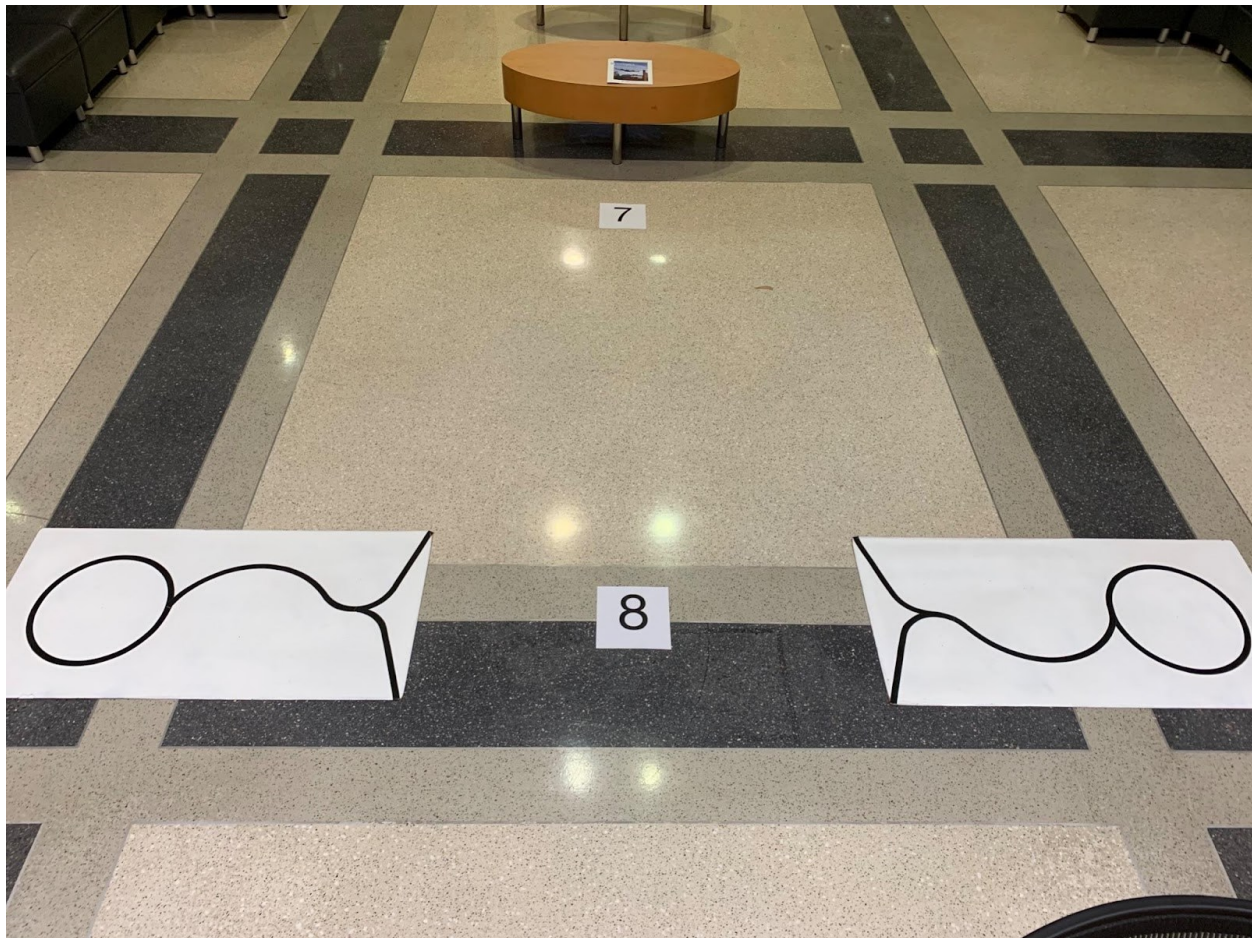


Figure 1.1 - Positions of pad 7, pad 8, and the two black line boards

Note: This is an approximate setup of the black line portion. I did not have the actual one-foot square mats for the picture, and the boards are actually usually placed further forward over top of the dark portion of the floor, so just use this figure to note the approximate *relative* positions of the setup.

Upon arriving within 2 feet of pad 8, a TA will start a timer. You will have 45 seconds to get 2 wheels on (or two wheels to pass over, which has the same effect) pad 8 and issue the blackline intercept command. You are to move from 7 to 8. Backing up to reposition is not permitted. You will not be able to move or give commands to re-position your vehicle as you approach 8. Have multiple commands for intercept that works with various alignments. Once two wheels have touched: you must **only execute one of your commands to intercept the black line**. Once issued, you **must place your controller on the ground** while your car is intercepting / traveling. No additional commands can be given until the vehicle has traveled around the black circle at least twice. At that time, a TA / Attendant **will instruct you to issue a *single* command to exit the circle and stop**. Your car must exit the circle and end more than 2 feet away from the black line circle and in a direction away from pad 8 (e.g., don't tell your car to exit and head back

in the direction of pad 8, as more students will be coming from that direction for their demo). It is permissible to complete more than two circles, so you can wait an entire circle after TA / Attendant gives you exit confirmation to send your command. That is the best way to ensure your exit direction is opposite pad 8.

Note: your car must intersect the black line at least six inches away from the circle. You are welcome to approach the board from any angle you would like, if it intersects the black *line* at least six inches away from the *circle*.

There are two boards with the line and circle design in the lab; to the left of the parts locker up against the wall. Be sure to test your line following code extensively on those boards, and make sure to bring them out to the atrium as well to test in the actual environment you will demo in. Do not leave the board out of the Lab.

While your car is traveling autonomously, the device you use to send commands (i.e. phone, computer, game controller, etc.) must be placed down on the ground or on a table to indicate that you are not sending commands. The only two commands that can be sent after two wheels have touched pad 8 are the command for your car to go autonomous, and the command for your car to exit the circle.

When your car is in autonomous mode, there are several messages that must be displayed on the top line of the display, and you must pause for at least 10 to 20 seconds between each of these events, giving the TA / Attendant individual a chance to check the display:

- **“BL Start” - while your car is traveling to the Black Line**
- **“Intercept” - when your car has intercepted the line.**
- **“BL Turn” - while your car is turning.**
- **“BL Travel” - while your car is traveling along the black line.**
- **“BL Circle” - while your car is traveling around the circle***
- **“BL Exit” - while your car is exiting the circle.**
- **“BL Stop” - once your car has stopped and successfully finished the course.**

* It is not necessary that the “BL Circle” be displayed *as soon* as you enter the circle, it purely must be displayed before you travel around the circle one full time. Often this is based on a timer from the turn. A pause of 10 to 20 seconds in the circle travel should occur for the TA / Attendant to see the display change.

Additionally, your car must stop and wait for at least 10 to 20 seconds between each of these events. That means, for example when your car first detects the black line, set the top line of the display to “Intercept” and wait for 10 to 20 seconds before making the turn. This will allow the TA / Attendant to confirm the car has indeed detected the black line. It also must stop for 10 to 20 seconds again after completing the turn, and once again when it realizes it is traveling around the circle. As noted above, this 10 to 20 second pause between “BL Travel” and “BL Circle” does not need to happen as soon as your car enters the circle, it just must happen within one rotation of entering the circle.

Display Messages

There are a lot of required messages for this project, so this section gives more details about what should be displayed during the different sections of the demo.

After Successful connection to Wifi, the IP Address must be placed on the display. Then, you can perform your calibration and display whatever you would like on the screen during calibration, but after calibration your display should look like the following. (The bars on either side should not be placed on your display, they are purely used to indicate in this document where the edges of the display are.)

Waiting		- The display must show “Waiting for input”
for input		with spacing of your choice
10.155		- IP address is not required at this stage but is recommended since
.102.55		you will want to be able to reconnect to your car if necessary

Once you switch to remote operation of your vehicle, you must display four pieces of information: the IP address or your Name, the seconds counter, the last command received, and “Arrived 0X” (when on a pad only). This is what your display should look like during IOT course operation:

Arrived 0X	- ‘X’ is a number 1-7. Should be blank when car is not on a pad.
10.155	- IP address with alignment and spacing of your choice.
.102.202	- You Can Use your Name on the 2 lines instead of the IP address.
F0001 999s	- Last received command and second’s counter.

You will only be required to display the five most-relevant characters of the last received commands (e.g., don’t display your passcode, as that is both redundant and a security concern). If your command structure has more than five characters of relevant information, use your discretion to display the five characters most useful to describe the operation, or what is easiest for you to display.

After you send the command that tells your car to go autonomous, the display does not change requirements, except that the top line will instead be used for the black line event descriptions instead of the arrived descriptions.

BL Start		- Current black line command, detailed above.
10.155		- IP address with alignment and spacing of your choice.
.102.202		- You Can Use your Name on the 2 lines instead of the IP address.
F0001 999s		- Last received command and second’s counter.

When you stop at the very end of the course, you must display “BL Stop”, a custom message of your choice indicating completion, and the total number of seconds the course took you.

BL Stop	- "BL Stop" command
That was	- Custom message to indicate completion.
easy!! ;-)	This is a major cause for celebration!!
Time: 999s	- Total time in seconds

Software

You may use any part of the previous code that you have developed, and no magic numbers are allowed. Headers for functions and files are required.

After Project 10, you will each submit your entire code base to Moodle, so make sure you look through it to add comment headers for functions and files, and that you have no more remaining magic numbers.

In order to submit your code to Moodle, first delete the contents of the DEBUG\LST and the DEBUG\OBJ files. Then, create a .zip file of your code and submit this compressed file to the submission window.

Project Write-up:

Complete the last project write-up for Project 10. Since this is the final write-up, every section is required. That means you will need to write a conclusion (this is you and your team's impression of the class and projects, overall), and also complete Section 5, which details how long your car would last with a fresh set of batteries.

The biggest change for Project 10 write-up is that the sections that call for code are required for the first time. You do not need flowcharts and the pasted code for *every* portion, just the sections outlined in previous project write-up instructions. Make sure that the code is well documented with comments.

Demonstration Procedure:

Demos will be in class during the last week of class.

You will place the car on or near the first pad with the display showing the correct information. Once the starter has acknowledged you are ready to go, start navigating the course.

The Rules:

- a. There is no overall time limit.
- b. Multiple students can start the course at the same time. There will be a starting area that will support multiple cars starting simultaneously.
- c. Multiple cars will be going on the course at the same time.
- d. Faster cars can go around / pass slower cars.
- e. You cannot touch your car after it leaves the starting area.
- f. If you need to touch your car, **you must return to the waiting line to start again.**

- g. Each arrival at a number must be accompanied by displaying the number on your display and **receiving confirmation from the TA** of acceptance.
- h. When you are arriving at pad 8, a timer may be started. Autonomous command must be given before the timer expires.
- i. Each of the lower black line elements must be displayed on the top line of the display: BL Start, Intercept, BL Turn, BL Travel, BL Circle, BL Exit, and BL Stop.
- j. At each Black line segment is completed, a **10 to 20 second** stop must occur before continuing.
- k. Re-tries at the start of the IOT course and before pad 7 will be allowed; pending students on their first attempt have precedence.
- l. Once you give the command to intercept the black line only the ending stop command can be given.
- m. Once the stop command is given, you have either made it to the end, or will have to start again before pad 7.
- n. Your car **MUST stop on its own after exiting the circle.**

Grading Scale for Project 10:

Demo 80%

<u>Item</u>	<u>Description</u>	<u>Points</u>
1	It completes the WiFi course	40
2	It displays the numbers identifying progression	5
3	It navigates to the black line	5
4	It stops at the black line	10
5	It turns and intercepts the black line	10
6	It travels the black line course	10
7	It navigates around the circle twice	5
8	It displays the black line progression	5
9	It exits the circle and stops	5
10	It stops	5
11	It waits 10 to 20 seconds between black line events	5
		105

Note the 5-point bonus for completing all action items.

In addition to this, if you demo both the WiFi course and black line portions on the first try, you will receive an additional 5-point bonus, so it is possible to score a maximum of 110 points on the demo portion.

Report 20%

<u>Item</u>	<u>Description</u>	<u>Points</u>
1	Project Write-up Notebook	<u>100</u>
		100

Appendix A - Miscellaneous Photos of Demo Setup:

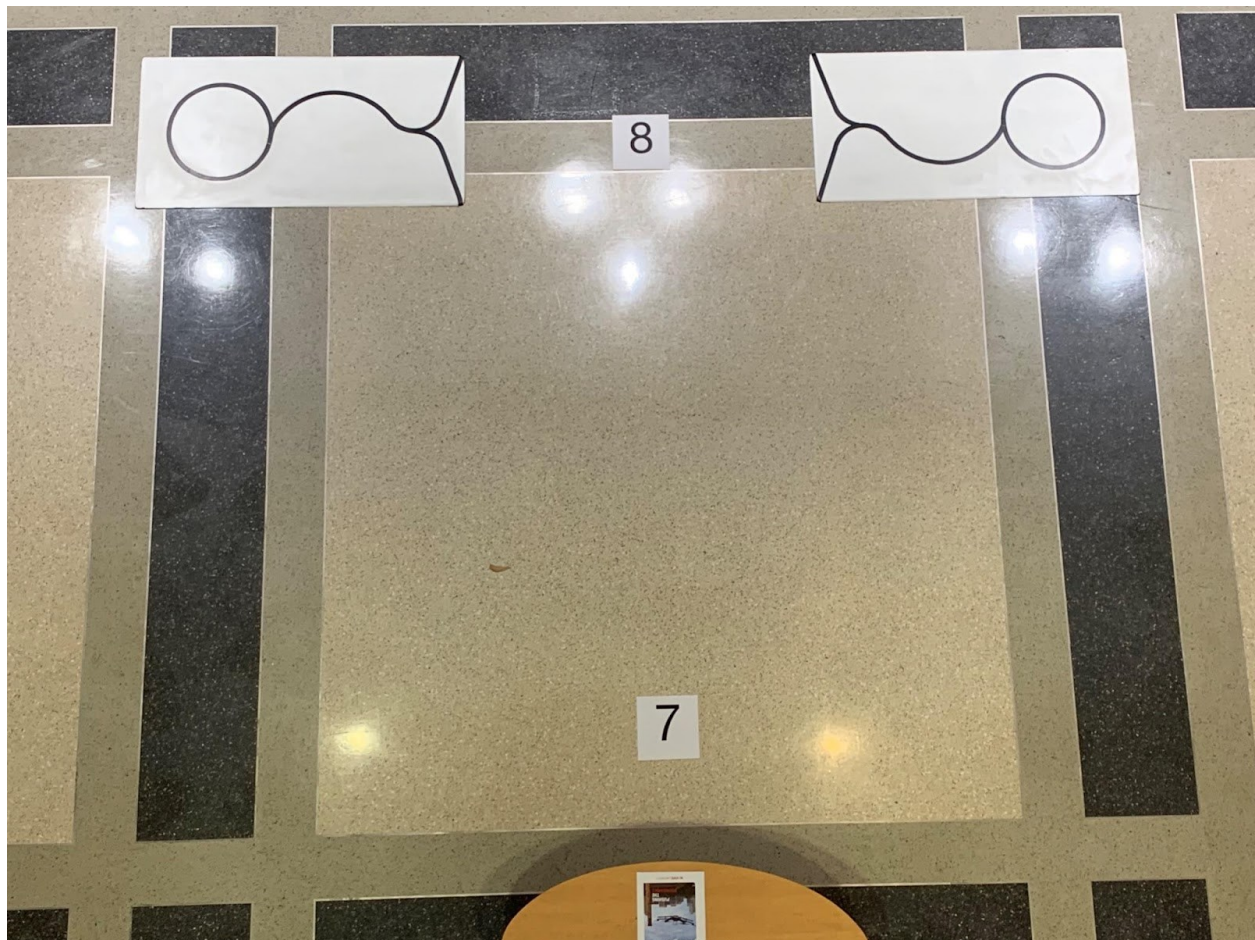


Figure A.2: Top view of black line portion.